simple pseudocode version for **Scrabble**:

1. Start the program.
2. Ask Player 1 for a word.
3. Ask Player 2 for a word.
4. Set both players' scores to 0.
5. For each letter in Player 1's word:
    - Convert the letter to uppercase.
    - Convert the letter into an array index.
    - Add the letter's point value to Player 1's score.
6. Repeat the same process for Player 2.
7. Compare the two scores.
8. If Player 1 has more points, print "Player 1 wins!"
9. If Player 2 has more points, print "Player 2 wins!"
10. If the scores are equal, print "Tie!"
11. End the program.

2. source code

```
#include <ctype.h>
#include <cs50.h>
#include <stdio.h>
#include <string.h>

// Points assigned to each letter of the alphabet.
// POINTS[0] is A, POINTS[1] is B, ..., POINTS[25] is Z.
int POINTS[] =
{
  1, 3, 3, 2, 1, 4, 2, 4, 1, 8,
  5, 1, 3, 1, 1, 3, 10, 1, 1, 1,
  1, 4, 4, 8, 4, 10
};

// Function prototype.
// compute_score() takes a word and returns its Scrabble score.
int compute_score(string word);

int main(void)
```

```
{
    // Get the words from both players.
    string word1 = get_string("Player 1: ");
    string word2 = get_string("Player 2: ");

    // Calculate the score for each player's word.
    int score1 = compute_score(word1);
    int score2 = compute_score(word2);

    // Compare the two scores.
    if (score1 > score2)
    {
        printf("Player 1 wins!\n");
    }
    else if (score2 > score1)
    {
        printf("Player 2 wins!\n");
    }
    else
    {
        printf("Tie!\n");
    }
}

// Compute the Scrabble score of a word.
int compute_score(string word)
```

```
{
    // Start the score at zero.
    int score = 0;

    // Go through every character in the word.
    for (int i = 0; i < strlen(word); i++)
    {
        // Ignore anything that isn't a letter.
        if (isalpha(word[i]))
        {
            // Convert the character to uppercase.
            char letter = toupper(word[i]);

            // Convert A-Z into an array index from 0-25.
            //
            // For example:
            // 'A' - 'A' = 0
            // 'B' - 'A' = 1
            // 'C' - 'A' = 2
            // ...
            // 'Z' - 'A' = 25
            int index = letter - 'A';

            // Add the letter's point value to the total.
            score += POINTS[index];
        }
```

```
    }


    // Return the final score.
    return score;
}
```

I turned a character into an array index by subtracting the ASCII value of `'A'` from the character's ASCII value. This makes `'A'` equal to index 0, `'B'` equal to index 1, and so on.